

A Probe of the Odd Parity Sector of the CvS Galerkin Matrix

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Status. Numerical report. No theorem is claimed and no claim is made about the Riemann hypothesis. Every newly reported numerical output is regenerated from harness/, except the **direct-quadrature completeness table in §1.3** — which no script produces and whose recipe is written out beside it — and the **derived $K(m)$ table in §5**, which is one division away from edge_precision.py's output, also written out. Four principal measurements and one follow-up correction are reported — the sector correspondence, the unseeded zero scan, the random-ensemble test of real-rootedness, and the T -scaling of the Galerkin exponent — each with its limits stated beside it. §6 records what the exchange with Groskin settled, and §7 states what remains open.

Novelty, stated plainly. Nothing here is proved. Every section reports a measurement, and the one thing that would matter theoretically — whether these roots converge to the zeta zeros as $c \rightarrow \infty$ — is exactly the open question this framework was built around, and is untouched by anything below. Whether the odd-sector behaviour is already known is not something this report could settle by itself; it was asked rather than assumed, and §6 records what came back. What is offered is a probe of a sector the published work lists as unprobed, with its controls and its limits attached.

What is and is not claimed, in one paragraph. The odd sector of the periodic family coincides exactly with a specific finite sine space. Its near-null vector yields a function whose **unseeded** sign-change scan of $(0.1, 100)$ returns twenty-nine sign-changing roots, each matching one of the twenty-nine zeta ordinates below 100, in order, with no additional sign-changing root detected — errors from 10^{-52} at γ_1 to 10^{-15} at γ_{29} — although the published CvS theorems assume the even sector. Random-ensemble experiments indicate that real-rootedness may extend to a global odd *kernel* vector, but not to the lowest vector of the odd sector as such. Nothing here is claimed to be new, proved, or explained.

Provenance of the numbers. Three different pieces of code produce what is below, and it matters which is which.

- My own fixed-support implementation is double precision and cannot speak to the deep eigenvalue regime; that was stated when this thread opened and is unchanged. It supplies exactly two things: the entrywise and bulk-spectral sector comparisons of §1.2 and the completeness table of §1.3.
- **Every matrix and every eigenvector in §2 comes from A. Groskin’s connes-cvs package in mpmath**, with a parity projection applied to the matrix it returns. The reconstruction F used for the unseeded scan is re-implemented from the published formula rather than called from the package — which is why §2 carries a cross-check against the package’s own `extract_zeros` on the same eigenvector.
- §3 uses neither: it is a synthetic ensemble drawn from CvS eq. (11) in numpy and mpmath, with no Weil form and no prime sum anywhere in it.

Attribution. The parity split of the sine modes, the exact parity decoupling, the edge expansion and the $(m-1)/(m+2)$ law are due to A. Groskin (issue #2 of `akivag613/connes-cvs-`). The answer that $L = u = \log c$ is structural for the CvS path is his. So are, by correspondence of August 2026 and recorded in §4.1 and §6: the provenance of Table 14, the withdrawal of the §8.2 mechanism and the second defect found in rewriting it; the cosh/sinh sign structure that closes the odd-dictionary question; and the statement that the even-sector real-rootedness mechanism is a parity-specific Toeplitz positivity argument with no odd analogue in print — all recorded in §6, with the sign structure carried in §3. The constructive form of that structure is B. Andrade’s, [10.5281/zenodo.20710075](https://zenodo.org/record/20710075). CvS Theorem 5.6, Theorem 6.1, eq. (11), eq. (21) and Appendix B.3 are Connes and van Suijlekom’s, arXiv:2511.23257.

1. The correspondence between the two constructions

1.1 The parameter map

`build_galerkin_matrix` sets $L = \log c$ and sums prime powers $n \leq c$; `extract_zeros` integrates against $\exp(2\pi i k x / L)$ on $[0, L]$, i.e. the basis $U_n(x) = L^{-1/2} \exp(2\pi i n x / L)$ of CvS Proposition 4.1. Against the sine basis

$$\varphi_j(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{j\pi(x + L/2)}{L}\right), \quad j = 1, \dots, m,$$

on $x \in [-L/2, L/2]$, this fixes

$$L = \log c, \quad p_c = c, \quad m = 2N,$$

which is the saturated diagonal. Everything below sits on it.

1.2 The odd sector: an exact identification on a subspace

The statement that the sine space is not an invertible basis change of the finite periodic Fourier space is correct as a statement about the whole spaces, and is not disputed here. What holds is narrower, and the variable matters: the periodic modes U_n live on $[0, L]$, so both sides are written in $x' = x + L/2 \in [0, L]$, in which $\varphi_j(x') = \sqrt{2/L} \sin(j\pi x' / L)$. Then

$$\frac{U_k - U_{-k}}{i\sqrt{2}} = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi k x'}{L}\right) = \varphi_{2k}(x'),$$

with no sign. The real projected combination used numerically, $(e_k - e_{-k})/\sqrt{2}$, differs from φ_{2k} by the overall phase i ; that leaves the Gram matrix and its zeros unchanged, and is noted only because the sine coefficients are imaginary in the complex-exponential basis. (Written instead in the centred variable x one has $\varphi_{2k}(x) = (-1)^k \sqrt{2/L} \sin(2\pi kx/L)$, so mixing the two variables introduces a spurious $\text{diag}((-1)^k)$; the algebra fixes the pairing used below.) **The two finite odd basis spaces coincide exactly. The observed spectral discrepancy decreases rapidly with T , consistently with archimedean truncation, though I have not established that this is its only source.**

Measured two ways, and the distinction matters. **Entrywise**, with G the periodic matrix restricted to its odd sector and S the sine matrix restricted to $\varphi_2, \varphi_4, \dots, \varphi_{2N}$, `compare_families.py` reports $\max_{i,j} |G_{ij} - S_{ij}| / \max |G|$ at $c = 13$, $N = 12$, $\text{dps} = 60$:

$$2.04 \times 10^{-4} \longrightarrow 2.87 \times 10^{-5} \quad (T = 100, 200),$$

falling with the archimedean truncation. **Two deliberately mispaired controls on the same quantity** show that the small figure is an identification and not a coincidence: pairing the periodic odd sector with the *odd*-indexed sine modes gives 0.856, and conjugating the correct pairing by $D = \text{diag}((-1)^k)$ — the sign pattern that appears if the two sides are read in different variables — gives 0.765. Both are of order one; the correct pairing is four orders below them and falling.

Spectrally, the same script reports the bulk metric

$$\Delta_{\text{odd}} = \frac{\max_{j \text{ in the largest six}} |\lambda_j(G) - \lambda_j(S)|}{\max(|\lambda_{\max}(G)|, |\lambda_{\max}(S)|)},$$

which at $c = 13$, $N = 16$, $\text{dps} = 80$ falls $1.71 \times 10^{-4} \rightarrow 2.42 \times 10^{-5} \rightarrow 3.22 \times 10^{-6}$ at $T = 100, 200, 400$. That is a comparison of the top six eigenvalues — neither an entrywise comparison nor a statement about the near-null eigenvalue. **I do not know whether either residual is entirely explained by the archimedean truncation.**

1.3 The even sector: completeness without a finite identification

The odd-indexed modes satisfy $\varphi_j(L - x') = (-1)^{j+1} \varphi_j(x')$, so for j odd they carry the same parity as $\{1, \cos(2\pi kx'/L)\}$ and are complete in the same subspace of $L^2[0, L]$ — but only asymptotically. Expanding the constant and $\cos(2\pi kx'/L)$ in them gives coefficients decaying like $1/j$ and an L^2 residual like $1/m$:

modes up to j	10	50	200	400
residual, constant	4.04×10^{-2}	8.10×10^{-3}	2.03×10^{-3}	1.01×10^{-3}
residual, $\cos(2\pi x'/L)$	8.30×10^{-2}	1.62×10^{-2}	4.05×10^{-3}	2.03×10^{-3}

No script in harness/ produces this table; it is a direct quadrature, quoted here so that it is reproducible without one. With $c_j = \int_0^L f \varphi_j$ over odd j on a uniform grid, the entry is $1 - \sum_{j \leq M} c_j^2 / \|f\|^2$; the values above are at $c = 13$ on 4×10^5 points, and the leading coefficients of the constant go 1.4419, 0.4806, 0.2884, 0.2060, 0.1602 at $j = 1, 3, 5, 7, 9$, i.e. $1.442/j$.

So the finite even blocks of the two constructions disagree. In the same top-six spectral metric, that disagreement was measured against T and is flat there — 5.0×10^{-2} , 4.7×10^{-2} , 4.9×10^{-2} at $T = 100, 200, 400$ — while the completeness above is a statement about the basis and not about the matrices. **That the block disagreement converges in m is therefore an inference from completeness, not something measured here**; the m -dependence of the even-block discrepancy was not swept. A naive $1/m$ extrapolation to a 10^{-55} function-space residual would require m of order 10^{55} ; **that is an extrapolation of an L^2 residual and not a conditioning bound on the matrix or its eigenvalues**, and no claim is made that the two quantities are related by that rate. The practical consequence is only that this route was not attempted.

2. An unseeded scan of the odd-sector function

Groskin's §10 lists among its limitations: “We compute the even sector only. The odd-sector zero, if any, is not probed.” This section probes it.

Protocol, stated before the numbers. The test function $F(\tau)$ is built from the odd-sector eigenvector coefficients alone. The interval $(0.1, 50)$ is then swept at step 0.02 for **sign changes**; each is refined by **bisection only**; the resulting root list is **written to disk**; and only after that is `mp.zetazero` called, to compare. No zeta ordinate enters the discovery at any point. (This differs from `extract_zeros`, which brackets `findroot` on $(\gamma_k - 0.005, \gamma_k + 0.005)$ and so is seeded at the answer; the two protocols agree where they overlap, see below.)

Four properties of `odd_sector_unseeded_scan.py` are checkable in the source rather than taken on trust: `scan_sign_changing_roots` admits a candidate only where two adjacent grid values differ in sign; `refine_bracket` is bracket-preserving bisection with $\lceil (\text{dps} + 8) \log_2 10 \rceil$ halvings and no Newton or secant step; the root list is written by `atomic_json` **before** `compare_after_discovery` is reached, and the only call to `mp.zetazero` in the file is inside that later function; and the residual recorded with each root is $|F|$ evaluated at the located point. The near-pole cutoff in `centred_transform` is $10^{-(\text{dps}-10)}$ rather than the package's fixed 10^{-130} — the cross-check below covers that difference.

Result at $c = 13$, $T = 400$, $\text{dps} = 120$. The scan returns **exactly ten roots**, and they are the first ten zeta ordinates **in order** — no extra sign-changing root in the interval, and none of the ten missing. Errors at $N = 100$:

k	1	2	3	4	5	6	7	8	9	10
$ r_k - \gamma_k $	8.70e-5	6.67e-5	4.28e-4	2.50e-4	1.14e-4	4.70e-4	2.74e-4	3.93e-3	1.10e-3	9.87e-36

and $\log_{10}$ of the first-zero error falls with N as -32.9 , -41.3 , -46.9 , -51.7 , -52.1 at $N = 20, 30, 40, 60, 100$.

Extended to height 100. Rerunning the sweep over $(0.1, 100)$ at the same step, on the cached $N = 100$ eigenvector from the $T = 600$ ladder rather than the $T = 400$ one above — which is why γ_1 reads 1.065×10^{-52} here against 8.70×10^{-53} in the table, the same T -shift described below — returns **twenty-nine sign changes, and there are exactly twenty-nine zeta ordinates below 100**. None missing, none claimed by two roots, and no spurious sign change anywhere in the interval. The errors continue smoothly: 1.065×10^{-52} at γ_1 , 1.199×10^{-35} at γ_{10} , 4.161×10^{-23} at γ_{20} , 2.711×10^{-15} at $\gamma_{29} = 98.83$.

The height law. Fitting $\log_{10}$ of the error against the ordinate over all twenty-nine gives

$$\log_{10}|r_k - \gamma_k| \approx 0.453\gamma_k - 58.0,$$

i.e. **about 0.453 decades lost per unit of height**, so at this cutoff and basis size the error reaches $O(1)$ near $\gamma \approx 128$. The fit is a straight line through points spanning 37 decades, and its residuals reach a few tenths of a decade at either end, so the crossing point is an extrapolation and not a measurement.

(The height-law paragraph below quotes 0.482 and $\gamma \approx 123$ for the same cutoff. That fit uses the ten zeros below 50 on the $T = 400$ vector; this one uses all twenty-nine below 100 on the $T = 600$ vector. The slopes differ by 6.4% and the $O(1)$ crossings by 4.1% — the disagreement between a ten-point and a twenty-nine-point fit of one line, not two different laws. The longer baseline is the better estimate.) That is the honest statement of the range: the agreement is not unlimited, it degrades linearly in the height, and it runs out a little past 100.

Consistency with the seeded route. At $N = 100$ the seeded `extract_zeros` gives 8.70e-53, 6.67e-50, 4.28e-48 for the first three — the same values. The two protocols agree; the unseeded one simply does not need the answer to find them.

The random control. A random odd vector in the same basis does produce sign-changing roots in the interval -9 to 14 of them across the $c = 13$ runs, and 11 at $c = 17$. In the $c = 13$, $N = 100$, $T = 600$ run its **closest approach to any zeta ordinate is 0.144**, against 1.199×10^{-35} for the worst of the ten in that same run — the two figures must come from one cell, and these do. Thirty-four orders of magnitude separate them, so the agreement is not a generic feature of the tested random odd controls.

That 0.144 is a single draw, and a single draw is the wrong form for this control. It is reproducible only from the same generator consuming its stream in the same order, not from the seed alone. Groskin, reimplementing the protocol independently, ran the control over five draws and obtained closest approaches of 0.104, 0.182, 0.187, 1.074, 0.120 (correspondence, August 2026); the value above sits inside that spread. **The claim the control supports is the separation and not the number.**

Independent reimplementation. Beyond the controls above, the campaign has since been reproduced by Groskin against the tagged release **without running the scripts of harness/** — the protocol reimplemented from its description, so that the agreement is independent evidence rather than a re-execution of this code (correspondence, August 2026). Every row of the constants table below reproduces in both sectors to the digits reported there; the ten-root and twenty-nine-root counts reproduce; and the worst of the ten at $c = 13$, $N = 100$, $T = 600$ comes out 1.19901×10^{-35} against the 1.199×10^{-35} reported here. Two anchors in that run are external to both of us: the $c = 13$, $N = 100$, $T = 400$ even sector returns $\lambda_{\min} = 2.0769626582 \times 10^{-59}$ and $|\gamma_1 \text{ err}| = 1.45495 \times 10^{-55}$, reproducing the published Table 2 values to their printed precision; and the published $T = 800$ value at $c = 17$ is 8047.4062, which is what the 8047 against 8052 comparison below rests on.

Cross-check against the package's own root finder. The reconstruction F above is re-implemented here from the published formula, so an error in that re-implementation would move every root and nothing else in this pipeline would notice. As a control, the same twenty-nine roots were recomputed with `extract_zeros` from the package itself, acting on the same cached eigenvector — an independent evaluation of the same mathematical reconstruction F , using a different solver (`findroot/Anderson` rather than `bisection`), and a bracket seeded at `mp.zetazero`. **Every one of the twenty-nine error figures agrees to all printed digits, and the root positions differ by between 10^{-62} and 4.4×10^{-59} — six orders below the smallest error in the table, which rules out the root-location implementation as the source of the observed errors. Seeding therefore changes where one looks, not what is found.**

Step control. The sweep was repeated at $c = 13$, $N = 60$, $T = 400$, $\text{dps} = 180$ with the step halved to 0.01: the same ten roots, and $\lambda_{\min}^{\text{odd}} = 5.07520120029 \times 10^{-55}$ identical to every

printed digit. So the list is not an artefact of the sampling interval at that resolution.

Three further cutoffs. The same unseeded protocol was run at $c = 17$ ($T = 400$, $\text{dps} = 180$, $N = 60, 80, 100$), at $c = 19$ ($\text{dps} = 220$, $N = 80, 100, 120$) and at $c = 23$ ($\text{dps} = 300$, $N = 100, 120, 140$). **All nine runs return ten roots, each closest to its own γ_k , sorted, none missing.** The random control returned 11, 11, 14 at $c = 17$; 11, 15, 12 at $c = 19$; 15, 13, 14 at $c = 23$ — again unstable for random vectors and stable for the eigenvector. At $c = 17$, $N = 60$ the errors are

$$1.240 \times 10^{-67}, 1.227 \times 10^{-64}, 9.547 \times 10^{-63}, \dots, 1.504 \times 10^{-50}, 1.633 \times 10^{-49}.$$

and the eigenvalue at each rung reproduces the earlier sweep to every printed digit — $2.41746920332 \times 10^{-70}$, $1.98674205805 \times 10^{-75}$, $2.13985905178 \times 10^{-76}$ at $N = 60, 80, 100$, against 2.41747×10^{-70} , $1.9867421 \times 10^{-75}$, $2.1398591 \times 10^{-76}$ from the earlier run at $\text{dps} = 80$.

What that agreement does and does not test. $\lambda_{\min}^{\text{odd}}$ comes from building the matrix and diagonalising the odd block; **no root finder touches it**, so the agreement says nothing about seeded against unseeded search. What it does establish at $c = 17$ is that the build, the parity projection and the eigensolver reproduce between the $\text{dps} = 80$ and higher-precision runs — useful, and a different question. The protocol itself is tested by the random control, the step control, the completeness count, and the cross-check against `extract_zeros` on a fixed eigenvector; those four are what carry it, not this. At $c = 19$ and $c = 23$ the eigenvalues likewise reproduce the earlier precision-resolved sweep; the contaminated $c = 19$, $N = 80$, $\text{dps} = 80$ cell is excluded as documented in §2.2. The reproduced values are $2.57039874544 \times 10^{-83}$ and $4.92683492454 \times 10^{-87}$ at $N = 80, 120$, and $6.42778802803 \times 10^{-103}$ and $7.43680210541 \times 10^{-107}$ at $N = 100, 120$.

The error tracks the eigenvalue, with a constant that grows in c . Writing

$$C = \frac{|\hat{\gamma}_1 - \gamma_1|}{\lambda_{\min}},$$

each read at the deepest common rung used for the paired comparison:

c	N (both sectors)	C_{odd}	C_{even}	ratio
13	100	432.425	7005.193	16.200
17	100	489.384	8052.033	16.453
19	120	510.275	8441.326	16.543
23	120	545.112	9097.646	16.689

Every row is one cell: the same c , the same N , the same $T = 400$ for both sectors. An earlier draft mixed the deepest available rung per sector, which makes a ratio of two numbers that were not measured together.

The $c = 23$ row is at $N = 120$ rather than deeper for a reason worth stating. The unseeded campaign reaches $N = 140$ in the odd sector, where $C_{\text{odd}} = 542.9$, but it computes no even block there. A separate convergence sweep does compute both sectors in one $N = 160$ cell and gives $C_{\text{odd}} = 542.7$, $C_{\text{even}} = 9048$ and a ratio of 16.671 — so $N = 120$ is the deepest paired cell whose odd value belongs to the unseeded campaign described in §2, and the $N = 160$ figures come from the seeded sweep instead. Both are reported; neither is mixed into the other.

The N -dependence must be reported with C , and the fourth cutoff shows why. At $c = 23$, C_{odd} runs 553.7, 545.1, 542.9 at $N = 100, 120, 140$ — so a table taken uniformly at $N = 100$

would put $c = 23$ two percent high while the other three are already settled there. At $c = 19$ it runs 520.0, 511.2, 510.3 at $N = 80, 100, 120$; at $c = 17$, 512.8, 491.8, 489.4 in the odd sector and 8514, 8103, 8052, 8049 in the even one at $N = 60, 80, 100, 120$, so the $N = 60$ figures overstate both by four to five percent. The even-sector value at $c = 17$ sits beside the published 8047 at the same cutoff, measured there at $T = 800$ against $T = 400$ here.

Fitting the four:

$$C_{\text{odd}} \approx 197.9 \log c - 73.7 \quad (R^2 = 0.9982), \quad C_{\text{even}} \approx 3673.6 \log c - 2392.3 \quad (R^2 = 0.9987),$$

the same functional form as the published even-sector fit $C \approx 6730 \log c - 11268$ (fifteen cutoffs, $R^2 = 0.956$), with different coefficients. Four points fit two parameters, so neither R^2 is evidence of much; the form is quoted because it matches, not because four points establish it.

The ratio varies substantially less than either constant over the four tested cutoffs. Across $c = 13$ to 23 the odd constant moves 26.1% and the even one 29.9% — they do not even grow at the same rate — while their ratio moves 3.0%, monotonically: 16.200, 16.453, 16.543, 16.689. Four points is four points.

C_{odd} is also stable in N from $N = 60$ onward at $c = 13$ (806.4, 534.0, 463.1, 433.1, 432.2, 432.2 at $N = 20, 30, 40, 60, 80, 100$, all at $T = 600$), and it **was stable under the one change of T that was tested**: at $c = 13$, $N = 60$ it is 433.4 at $T = 400$ and 433.1 at $T = 600$, although λ itself moves 24% between those two. That is a single test at a single cell, not a demonstration of T -independence. What it does show is that at that cell the absolute error is not T -converged because λ is not, while the ratio is.

Height law across the cutoffs. Fitting $\log_{10}$ of the error against the ordinate at $N = 100$: 0.482, 0.504, 0.526 **decades lost per unit of γ at $c = 13, 17, 23$** , with the error reaching $O(1)$ near $\gamma \approx 123, 160, 204$. The slope creeps up by about 9% across that range while the reach grows by 66%: **in these three fits, most of the observed gain is in the reachable height rather than in an improvement of the decay rate**. At $N = 60$ the slopes are almost the same (0.483 and 0.513 at $c = 13, 17$), so the observed trend does not appear to be driven primarily by N on the rows tested.

Multi-zero behaviour across four cutoffs. With $c = 13, 17, 19, 23$ all at $N = 100$, the per-zero slope is a fit with two degrees of freedom. Regressing $\log_{10} |r_k - \gamma_k|$ on $\log c$:

Since the error falls as c grows, the fitted slope is negative; the positive quantity below is the **decay exponent** $\alpha_k = -d \log_{10} |r_k - \gamma_k| / d \log c$.

k	1	2	3	4	5	6	7	8	9	10
α_k	83.14	82.86	82.65	82.30	82.12	81.74	81.44	81.20	80.69	80.49
ratio to $k = 1$	1.000	0.997	0.994	0.990	0.988	0.983	0.980	0.977	0.971	0.968

with mean $R^2 = 0.9989$. **The spread across the ten is 3.2%**, against the published even-sector figure of all ten within 3.8% over fifteen cutoffs. Read on two cutoffs the same statistic gave 3.6%, on three 3.4%: **the spread decreases modestly as points are added, an encouraging but still short sequence**.

Two checks on that number. The $c = 19$ errors are not converged at $N = 100$ — going to $N = 120$ moves them by a factor 3.1 — but that is a nearly uniform shift in $\log$ across all ten k , so it moves the absolute slopes (to 82.16... 79.46) and leaves the **spread at 3.3%**. And read at $N = 60$, where none of the cutoffs has settled, the same statistic gives 6.9% — nearly

double. The spread is robust to the first and not to the second, which is worth stating rather than quoting one figure.

Four cutoffs are still four, not fifteen, and the fit has two degrees of freedom.

What is not established. Four cutoffs only, $c = 13, 17, 19, 23$; heights below 100 at $c = 13$ and below 50 at the other three. The scan detects **sign changes**, so a root of even order would be invisible to it, and two roots closer together than the step would merge — halving the step to 0.01 leaves the list unchanged, which bounds that risk at this resolution but does not exclude it in general. The interval begins at 0.1, so anything near the origin is excluded by construction. Nothing here is a theorem, and none of it bears on the Riemann hypothesis.

And $\lambda_{\min}^{\text{odd}}$ itself is **not converged in T** : at $\text{dps} = 180$ it is identical to the $\text{dps} = 120$ value to every printed digit, but moving $T = 400 \rightarrow 600$ shifts it by +11.7%, +7.3%, +7.3%, +24.1%, +23.0%, +22.5% at $N = 20, 30, 40, 60, 80, 100$. So at the 10^{-55} level the eigenvalue sits on a T -limited plateau rather than an N -limited one. The root positions move with it: at $c = 13$, $N = 60$ every one of the ten errors grows by the same factor 1.24 that λ does, which is why the ratio above is the stable quantity and the absolute error is not.

2.1 Controls

Four vectors from each sector, all from one cell — $c = 13$, $N = 60$, $T = 400$, $\text{dps} = 80$ — put through the **seeded** `extract_zeros`, which is the cheaper route when the question is only whether a vector reconstructs at all. The near-null figures are therefore the $N = 60$ ones and not the deeper $N = 100$ values quoted above.

vector	$ \hat{\gamma}_1 - \gamma_1 $, even sector	$ \hat{\gamma}_1 - \gamma_1 $, odd sector
near-null	4.22×10^{-55}	2.20×10^{-52}
largest eigenvalue	1.27	0.78
random (Gaussian coefficients)	did not converge in the seeded search	0.34
second smallest eigenvalue	1.32×10^{-49}	7.35×10^{-47}

The even random vector did not converge to a root in the seeded search, while the odd random vector returned one 0.34 away; a failure of `findroot` does not establish that no root exists in the window. **Close reconstruction occurred at the low end of the spectrum, not only at the very bottom:** the second-lowest vector also lands within 1.32×10^{-49} and 7.35×10^{-47} , five to six orders worse than the ground vector but nothing like the controls. The largest-eigenvalue and random vectors did not closely reconstruct in these trials. That is an observation over four vectors per sector, not a law: no functional relation between the eigenvalue and the root error is claimed here, and establishing one would need a sweep over the whole low end of the spectrum with λ tabulated beside the error.

2.2 The lag between the sectors

Every cell run, at $T = 400$. The lag is $\log_{10}$ of the ratio of the two first-zero errors.

c	N	dps	$\lambda_{\min}^{\text{even}}$	$ \hat{\gamma}_1 - \gamma_1 _{\text{even}}$	$ \hat{\gamma}_1 - \gamma_1 _{\text{odd}}$	lag
13	40	80	6.79298×10^{-54}	5.17506×10^{-50}	1.19574×10^{-47}	2.364
13	60	80	5.99929×10^{-59}	4.21519×10^{-55}	2.19967×10^{-52}	2.718

c	N	dps	$\lambda_{\min}^{\text{even}}$	$ \hat{\gamma}_1 - \gamma_1 _{\text{even}}$	$ \hat{\gamma}_1 - \gamma_1 _{\text{odd}}$	lag
13	100	80	2.077×10^{-59}	1.4550×10^{-55}	8.6968×10^{-53}	2.776
17	60	80	3.52427×10^{-74}	3.00057×10^{-70}	1.23960×10^{-67}	2.616
17	60	150	3.52427×10^{-74}	3.00057×10^{-70}	1.23960×10^{-67}	2.616
17	80	150	1.71325×10^{-79}	1.38825×10^{-75}	9.77031×10^{-73}	2.847
17	100	250	1.40108×10^{-80}	1.12815×10^{-76}	1.04721×10^{-73}	2.968
17	120	250	1.01444×10^{-80}	8.16509×10^{-77}	7.80056×10^{-74}	2.980
19	60	80	1.14409×10^{-79}	1.04719×10^{-75}	4.54572×10^{-73}	2.638
19	80	80	6.89173×10^{-82}	1.00758×10^{-78}	1.23528×10^{-79}	-0.912 (discarded)
19	80	200	2.02675×10^{-87}	1.75041×10^{-83}	1.33669×10^{-80}	2.883
19	120	250	2.42209×10^{-91}	2.04456×10^{-87}	2.51404×10^{-84}	3.090
23	60	150	2.76142×10^{-88}	3.04435×10^{-84}	1.18794×10^{-81}	2.591
23	80	200	1.14292×10^{-99}	1.11320×10^{-95}	8.04765×10^{-93}	2.859
23	100	250	4.18176×10^{-107}	3.87555×10^{-103}	3.55935×10^{-100}	2.963
23	120	300	3.13658×10^{-111}	2.85355×10^{-107}	4.05389×10^{-104}	3.152
23	160	350	5.83185×10^{-113}	5.27672×10^{-109}	9.60204×10^{-106}	3.260

Bold rows are the deepest N reached at each of $c = 13$ and $c = 17$; they are not equally settled, and the convergence paragraph below this table says how far each is.

Discarded row. At $c = 19$, $N = 80$, $\text{dps} = 80$ the eigenvalue depth is 82 digits and the two sectors returned 6.89×10^{-82} and 4.40×10^{-82} , reading as the odd channel winning. At $\text{dps} = 200$ the same cell gives 2.03×10^{-87} and 2.57×10^{-83} . This is the backward-error contamination analysed in Groskin §7, at a different N .

Precision control. The two $c = 17$, $N = 60$ rows are the same cell at $\text{dps} = 80$ and 150; they agree to every printed digit.

Convergence. The lag climbs with N at fixed c — 2.364, 2.718, 2.776 at $c = 13$; 2.616 through 2.980 at $c = 17$ — so no lag figure is meaningful until its N -step has settled. The last N -step at each cutoff moves the error by

$$2.9 (c = 13, N = 60 \rightarrow 100), \quad 1.38 (c = 17, 100 \rightarrow 120), \quad 8561 (c = 19, 80 \rightarrow 120), \quad 54 (c = 23, 120 \rightarrow 160)$$

and none of them is small enough to call the underlying error converged. The lag is the steadier quantity, and every cutoff does have a last step to read it on:

$$0.058 (c = 13), \quad 0.012 (c = 17), \quad 0.207 (c = 19), \quad 0.108 (c = 23)$$

decades over each cutoff's deepest available N -step ($2.718 \rightarrow 2.776$, $2.968 \rightarrow 2.980$, $2.883 \rightarrow 3.090$, $3.152 \rightarrow 3.260$). So $c = 17$ is much the steadiest and $c = 19$ and $c = 23$ the least, which matches the error steps above. **None of the four is treated here as a converged bound.** The lag increases monotonically with N in all four series measured here, which suggests the unsettled rows understate it, but no monotonicity in N is established and none of them is quoted as a bound.

Across the three intervals the even channel gains 5.31, 5.30, 5.40 decades per unit of c ; the lag grows by 0.051, 0.055, 0.043 per unit, about one percent of that. **Only the first of those three intervals rests on two of the better-settled rows;** the second and third use the $c = 19$ and $c = 23$ figures, which are provisional in the sense just described, so the near-constancy across the three should be read as suggestive rather than measured.

3. Is real-rootedness forced by the matrix class?

Evenness is a hypothesis in every statement this rests on: CvS Theorem 5.6 requires the one-dimensional kernel to be even with respect to γ ; Theorem 6.1 the lowest eigenfunction; CCM Theorem 1.1 the smallest eigenvector; and Groskin's finite Guinand-Weil dictionary is stated for real even Galerkin coefficient vectors. The vector of §2 satisfies none of them: the even sector is always lower, so it is the lowest eigenvector **of the odd sector**, and it is not in the kernel.

To see whether that matters, random matrices were drawn from the **algebraic matrix class of CvS eq. (11)** — an ensemble respecting that algebraic structure, not matrices arising from the Weil form itself. **The sampling law, stated in full so that the observation can be read for exactly what it covers:** the free parameters $a_0, \dots, a_N$ and $b_1, \dots, b_N$ are drawn i.i.d. standard normal with `numpy.random.default_rng(7)`, and the remaining entries are fixed by the symmetry below. The class is

$$q_{i,i} = a_i, \quad q_{i,j} = \frac{b_i - b_j}{i - j} \ (j \neq i), \quad a_{-i} = a_i, \quad b_{-i} = -b_i,$$

and $Q - \lambda_{\min} I$ formed — positive semidefinite of the same form, with the ground eigenvector in its kernel. The roots of eq. (21),

$$P(s) = \sum_{k \in \{-N, \dots, N\}} \xi_k \prod_{\substack{j \in \{-N, \dots, N\} \\ j \neq k}} (j - s),$$

were then tested. **The root test was validated against CvS Appendix B.3 first:** at $N = 2$ with $u = 1$ the roots are real exactly when $v \notin (-2, -\frac{1}{2})$, agreeing with their criterion at all 24 values of v tried.

Experiment A — the global kernel vector is odd. This is Appendix B.3's configuration, carried to higher N . Roots real in **693 of 693** trials at $N = 3, 4, 5, 6, 8$. The float64 failures are root-finding noise on the degree- $2N$ polynomial; recomputed in mpmath, **no nonzero imaginary part was detected at the working precision.**

Experiment B — the global ground state is even, and the vector tested is the lowest eigenvector of the odd sector. This is the configuration of §2. Roots real in **686 of 1307** trials, about 52%, with genuine failures reaching $|\text{Im}| \approx 1.7 \times 10^2$.

Both experiments re-test every float64 failure in mpmath before counting it: a degree- $2N$ polynomial is badly conditioned in double precision, and an apparent complex root there is not evidence of one. Without that recheck the raw float64 counts read 683/693 and 633/1307; the figures above are the corrected ones.

Both experiments, the mpmath recheck and the Appendix B.3 validation are in `harness/odd_vector_reality.py`; it refuses to report unless A comes out all-real and B lands between 30% and 70%, so a silent regression in either would fail rather than print.

Reading. Real-rootedness in configuration B is **not forced in the sampled ensemble**: close to half the draws break it, against none in configuration A. No explicit certified counterexample is exhibited here, so this is a statement about the ensemble and not a theorem about every matrix of the form. The CvS matrices possess some additional structure not captured by this random ensemble; **whether that structure is arithmetic is precisely what is not known here**. The ensemble was not designed to isolate any particular constraint, and a non-arithmetic constraint absent from it would produce the same contrast.

Recorded as a sampled-ensemble observation under the stated sampling law, not as a conjecture — the conservative of the two forms Groskin offered. He confirms that the even-sector mechanism in CvS runs through a **Toeplitz positivity argument that is parity-specific**, and that he knows of no odd analogue in print (correspondence, August 2026), so the contrast above is not answered by anything published that either of us is aware of.

A mechanism that bears on the parity asymmetry, though not on this experiment. The same correspondence points at Andrade, [10.5281/zenodo.20710075](https://doi.org/10.5281/zenodo.20710075), where the prime block is written as a Loewner divided-difference matrix and the sign of each prime-power term is fixed by the parity of the test vector through its Fourier symbol: real and positive for an even vector, so the terms add without cancellation, and purely imaginary and sign-changing for an odd one, so they compete and positivity survives only as a sum, with the measured transition near $q \sim \sqrt{c}$. The imaginary and sign-changing half of that follows from parity alone and therefore covers **any** real odd vector, not only the $\sinh$ he works with. **It is a candidate account of why the two sectors behave differently and it is not tested here**: his statement is about the sign of each per-prime term, the quantities measured above are root counts and reconstruction errors, and nothing yet connects the two.

4. The T -scaling of the Galerkin exponent

Groskin §8.2 proposes $s = \sigma_{\text{eff}} T / 2\pi$ and §11 asks for the N -convergence measurement to be repeated at larger T . Regressing $\log_{10} |\lambda_{\min}^{\text{even}}|$ on $\log_{10} N$ as in §8.1, at a fixed N grid, $T = 400$ against $T = 1600$:

c	N grid	$T = 400$	$T = 800$	$T = 1600$
13	{20, 25, 30}	24.23	—	24.14
23	{40, 60, 80}	46.14	46.03	45.93

with $R^2 \geq 0.9990$ on every fit. At $c = 23$ the three values were all measured here at $\text{dps} = 250$ on the published N grid, so they form a sequence rather than a pair: **each doubling of T moves s by 0.23% and then 0.21%**, and the quadrupling from 400 to 1600 by 0.46%. At $c = 13$ the quadrupling moves it by 0.37%. The mechanism $s = \sigma_{\text{eff}} T / 2\pi$ predicts that each doubling **doubles s** .

The $T = 400$ row at $c = 23$ is also a numerical reproduction in a separate higher-precision run: our $\log_{10} |\lambda| = -71.179, -87.559, -98.942$ against the published $6.628 \times 10^{-72}, 2.761 \times 10^{-88}, 1.143 \times 10^{-99}$ — the same three numbers to every printed digit, giving the same $s = 46.14$.

The reason is visible in the raw values: the T -shifts in $\log_{10} |\lambda|$ are 0.046-0.080 at $c = 13$ and 0.088-0.213 at $c = 23$ — nearly constant across each grid, so they move $\log \lambda$ almost uniformly and change the fitted slope against $\log N$ only slightly. They are not exactly constant, which is consistent with the residual movement in s .

One number the script prints must not be read as a comparison with the published fit: it reports a slope of s against c from the two cutoffs, 38.4 at $T = 400$ and 38.2 at $T = 1600$. Those mix N grids — $s(13)$ is on $\{20, 25, 30\}$ and $s(23)$ on $\{40, 60, 80\}$ — so they are not comparable with the published slope 55, because that fit uses six cutoffs measured on the common grid $\{40, 60, 80\}$ (refitting its six points gives $s(c) \approx 54.5 \log c - 127.6$). Only the T -ratio at a fixed c and a fixed grid is used above.

4.1 Which T Table 14 belongs to, and what the measurement settles

An earlier draft asked which T the published Table 14 was computed at, because §8.2 treats its slope 55 as a $T = 800$ datum ($A = 55 \cdot 2\pi/800 = 0.432$) while Table 8 — the $c = 23$ N -sweep the measurement rests on — is labelled $T = 400$.

Groskin has resolved it (correspondence, August 2026): Table 14's rows come from the per-cutoff N -convergence ladders at $T = 400$, $\text{dps} = 150$, **on** $N \in \{40, 60, 80\}$, and §8.2's $T = 800$ came from the sweep convention. That also explains why the $c = 23$ reproduction above is exact to every printed digit: **that table is its source**.

And on what the three- T measurement settles. He has since recomputed all nine cells independently, on the code released as `connes-cvs 0.3.0`, obtaining 46.14000, 46.03143, 45.93406 against the values above — agreeing to 4.1×10^{-3} at worst. **On that basis §8.2's mechanism is withdrawn:** in his words, the measurement is the §11 test executed, and it fails; the constant $A = 0.432$ has no meaning as derived. Proportionality in T would have put $s(1600)$ near 184.6 against a measured 45.93.

The withdrawal is his and is recorded here as his. What this report measured is narrower and is all it claims: that s is nearly flat in T at fixed c on a fixed grid. Table 4 of the paper already reports a 0.14 shift between $T = 400$ and 800 at $c = 13$, so the insensitivity begins before this test.

A second defect surfaced in the same rewrite, and it is his finding, not this report's: §8.2 reasoned as though the Galerkin method acted on $[0, T]$, when the form acts on $[0, L]$ with $L = \log c$ and T bounds only the archimedean integral — so the substitution producing $A = 0.432$ was wrong in two ways rather than merely fed the wrong cutoff.

Both corrections are public in the package's `ERRATA.md`, carried in the text of the revised manuscript, deposited as Zenodo Version 3.6 under the concept DOI [10.5281/zenodo.19546514](https://doi.org/10.5281/zenodo.19546514), and submitted to arXiv as a replacement for 2605.20224.

The comparison he specifies for a cross-cutoff slope, which this report does not yet carry: $N \in \{40, 60, 80\}$ at $T = 400$, $\text{dps} = 150$, with $c = 17$ and $c = 23$ — leaving $c = 13$ out, since it is already saturating on that grid, which is why its row carries $R^2 = 0.87$. `har-ness/sobolev_slope.py --stage1` runs exactly that grid.

Limits. Two cutoffs. At $c = 13$ the grid $\{20, 25, 30\}$ is not the published $\{40, 60, 80\}$, so the value $s = 24.23$ is not comparable with the published $s(13) = 9.4$; only the T -ratio at a fixed grid is, and that is what is reported. At $c = 23$ the grid is the published one and all three T were measured here, which is why that row carries the weight. What is not tested is whether the same flatness holds at cutoffs beyond 23, or on a grid deeper than $N = 80$.

A precision note. At $T = 1600$ with $\text{dps} = 80$ the cells returned $\log_{10} |\lambda| \approx -3.4$ with five negative eigenvalues, forty-four orders adrift, at a cutoff where the matrix is positive on every cell. At $\text{dps} = 200$ the same cell returns to -38.832 with none. The tell was the sign count,

not the magnitude: the failure made λ too shallow rather than too deep, so a floor test on digit count would not have caught it.

5. A follow-up on the $(m-1)/(m+2)$ discrepancy

The ε^2 scaling was reported earlier at $m = 20$. The prefactor depends on m ; writing

$$\text{discrepancy} = K(m)\varepsilon^2$$

— K to keep it distinct from the $C(m)$ of $\sigma_2/\sigma_1 = C(m)\varepsilon^4$ — the ratio between successive decades of ε is 100.00 at every m tested, and

m	10	20	40	60	80	100	150	200
$K(m)$	9.327	20.968	44.540	68.186	91.853	115.527	174.730	233.942

so $K(200)\varepsilon^2 = 2.339 \times 10^{-6}$ at $\varepsilon = 10^{-4}$, which is the figure 2.4×10^{-6} quoted in the thread. It is not a float64 floor, as had been assumed here.

`harness/edge_precision.py` prints the law itself — σ_2/σ_1 against $(m-1)/(m+2)$ for each (m, ε) — but not K . The table above is that output divided through: $K(m) = |\sigma_2/\sigma_1 - (m-1)/(m+2)|/\varepsilon^2$, evaluated at $\varepsilon = 10^{-4}$ and confirmed at 10^{-5} and 10^{-6} , where the decade ratio is 100.00 throughout. The parity-block shortcut — taking the top two $|\cdot|$ from the two blocks separately rather than from the full matrix — is what made $m = 150$ and 200 reachable; it was validated at $m = 40$, where both routes give 0.928571873973.

6. What the exchange settled

Four questions were put to Groskin. Three are now closed; the record is kept short here because in each case the substance lives where it belongs in the report.

question	outcome
Does the finite Guinand-Weil dictionary extend to odd v ?	No, and not as an editorial choice. The even statement rides on a cosh-type pairing whose prime-power terms carry one sign; the odd analogue is sinh-type with mixed signs and needs its own tail treatment. No obstruction of principle, but an open construction rather than a corollary. The constructive form of that sign structure is Andrade's — see §3. Nothing here claims the dictionary extends, and §2 does not use it.

question	outcome
A general real-rootedness theorem for an odd ground-state kernel vector?	Not one he knows of. The even-sector mechanism runs through a parity-specific Toeplitz positivity argument, with no odd analogue in print. §3 therefore records the ensemble contrast as a sampled-ensemble observation with its sampling law stated, not as a conjecture.
Which T does Table 14 belong to, and how should the §11 test be run?	Answered and published — $T = 400$, $\text{dps} = 150$, $N \in \{40, 60, 80\}$. The measurement of §4 is the §11 test executed; on its basis the §8.2 mechanism is withdrawn and a second defect in that section corrected. Both are in ERRATA.md and the revised manuscript. See §4.1.
The float L in <code>extract_zeros</code> .	Fixed and shipped in connes-cvs 0.3.0: the function takes c and computes L internally at the active precision; a float c is refused, a float L warns, and the quick-start is corrected. Reported independently at issue #3 .

For the record, since it is the reason the fourth item was raised at all rather than filed as caller error: passing $L = \text{math.log}(13)$ yields a first-zero error of 1.41×10^{-17} where $L = \text{mp.log}(13)$ yields $1.4549524 \times 10^{-55}$, with $\lambda_{\min}^{\text{even}} = 2.0769626582 \times 10^{-59}$ either way — and the quick-start example itself passed the float, so the 10^{-17} ceiling was the expected outcome of following the documentation. Separately, the `build_galerkin_matrix` docstring gives the basis as $\exp(2\pi i k t / (2 \log c))$ while the implementation uses $2\pi k / L$ with $L = \log c$, matching Groskin §2.2.

7. What remains open

Whether the odd-sector reconstruction has an explanation inside the theory. §2 reports that the scan finds the ordinates and nothing else, at four cutoffs and to height 100 at the smallest. Why a sector the dictionary does not cover should reconstruct at all is not addressed here, and the two answers above make it less likely that it is already known: the dictionary does not extend, and there is no odd analogue of the positivity mechanism in print.

Whether the §3 contrast reflects arithmetic structure. The sampled ensemble breaks real-rootedness in about half of configuration B and in none of configuration A. The CvS matrices carry some structure the ensemble does not — but the ensemble was not designed to isolate any particular constraint, and a non-arithmetic constraint absent from it would produce the same contrast. The parity mechanism noted in §3 is a candidate account of the asymmetry and is untested against these numbers.

Whether the flatness of §4 persists. The exponent is nearly insensitive to T at two cutoffs on two grids, and that measurement has already cost one mechanism. What it does not establish is the behaviour at cutoffs beyond 23, on grids deeper than $N = 80$, or under a reparameterisation of the mechanism that was not considered. The cross-cutoff slope on the grid specified in §4.1 has not yet been run.

And the standing one, which none of this touches. Whether the CvS roots converge to the zeta zeros as $c \rightarrow \infty$ is the question the construction exists to raise, and nothing in this report bears on it, or on RH.

8. Reproducing

All scripts in `harness/`. They require `pip install connes-cvs` and, for anything above $N \approx 40$, `python-flint`.

script	what it regenerates
<code>compare_families.py</code>	§1.2, the top-six bulk-spectral sector comparison, with gates
<code>odd_sector_unseeded_scan.py</code>	§2, the unseeded scan, the frozen root list and the random control
<code>odd_sector_scan_height100.py</code>	§2, the extension to height 100 from the cached eigenvector
<code>cross_check_extract_zeros.py</code>	§2, the two-implementation cross-check on all 29 roots
<code>odd_sector_zeros.py</code>	§2, the seeded cross-check against <code>extract_zeros</code>
<code>reconstruction_controls.py</code>	§2.1, the four control vectors
<code>convergence_sweep.py</code>	§2.2, the lag table
<code>odd_vector_reality.py</code>	§3, experiments A and B and the B.3 validation
<code>sobolev_slope.py</code>	§4, the T -scaling; <code>--stage1</code> runs the cross-cutoff grid of §4.1
<code>edge_precision.py</code>	§5, the $(m-1)/(m+2)$ law itself; $K(m)$ is one division away, written out in §5

AI assistance

The verification scripts and much of the prose in this paper were written with the assistance of Claude (Anthropic); the research direction, the decisions about what to publish, and responsibility for every claim are the author's. See the repository README for a fuller statement.